

Annamalai Kalyanasundaram

Software Developer | GenAI Specialization (In Progress) | AWS Solutions Architect

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SUMMARY

Software engineer with ~2 years delivering production cloud and API systems, currently completing a Generative and Agentic AI specialization (expected October 2026). Building specialization capstones in LangGraph/LangChain RAG and agentic workflows, with planned packaging as a containerized FastAPI service. Production background includes REST APIs, React, MySQL, AWS, Docker, and CI/CD. AWS Certified Solutions Architect – Associate.

TECHNICAL SKILLS

Languages: Python, JavaScript / Node.js, SQL, Bash, Kotlin

Backend & Frontend: Node.js, Express.js, RESTful APIs, React.js, MySQL

Cloud & DevOps: AWS (EC2, VPC, S3, IAM, Lambda, ECS, RDS, CloudWatch), Docker, Jenkins, GitHub Actions, Git, Azure AD

Mobile: Kotlin, Jetpack Compose, Android SDK

GenAI (in progress†): LangChain, LangGraph, CrewAI, RAG, vector databases (FAISS, ChromaDB, Pinecone, Milvus), RAGAS, LangSmith, Hugging Face, PyTorch, Transformers, prompt engineering

† Generative & Agentic AI specialization and capstones — expected completion October 2026. FastAPI and Kubernetes are learning targets for the lead capstone, not production experience.

EXPERIENCE

Software Developer

Jun 2024 – Jul 2026

Arche Global Pvt Ltd — Bengaluru, India

- Enhanced Pulse, a cloud operations platform for hybrid cloud environments, driving a ~30% improvement in operational efficiency through automated monitoring and streamlined workflows.
- Designed and maintained 15+ RESTful API endpoints (Node.js, Express.js) for asset management, ticketing, and monitoring under high-frequency read/write traffic.
- Established CI/CD workflows that reduced deployment time by streamlining release cycles and minimizing manual intervention across staging and production.
- Optimized MySQL database queries and schema design for high-frequency operations, improving data retrieval performance.
- Built responsive web interfaces in React.js for asset management, ticketing, and real-time cloud monitoring dashboards.
- Collaborated with cross-functional DevOps and infrastructure teams to deliver sprint commitments in an Agile environment.

GENAI SPECIALIZATION & CAPSTONES

Advanced Generative and Agentic AI

In progress · Expected Oct 2026

Naresh Technologies

- Foundations:** Python for AI, neural networks, PyTorch, Transformer architecture, Hugging Face, multimodal LLMs, prompt engineering and guardrails.
- RAG systems:** Chunking, embeddings, ChromaDB, Pinecone, FAISS, Milvus, conversational RAG, RAGAS evaluation, Corrective RAG, Self-RAG, hybrid search, GraphRAG.
- Agentic AI:** LangChain, CrewAI, LangGraph (state graphs, persistence, human-in-the-loop), MCP, A2A, LangSmith observability.
- Optimization:** PEFT fine-tuning (LoRA/QLoRA), quantization, vLLM (coursework).

Agentic RAG Advisor — Lead Capstone

In progress · Target Oct 2026

Planned stack: LangGraph · LangChain · ChromaDB / FAISS · RAGAS · LangSmith · FastAPI · Docker

- Designing a multi-agent advisor for a regulated domain: LangGraph orchestration with memory over a RAG pipeline (ingestion, chunking, embeddings, vector search, prompt assembly), with retrieval tuning, guardrails, RAGAS evaluation, and LangSmith tracing.
- Intended packaging as a containerized FastAPI service with REST endpoints, request/response validation, structured error handling, and basic observability.

Additional Capstones

In progress · Target Oct 2026

- Domain-aware LLM chatbot with switching between open-source (Mistral) and API models.
- RAG-powered customer support assistant over internal knowledge bases.
- Agentic market-insight generation using web search tools.

ADDITIONAL PROJECTS

BIAL Plus — Internal Employee Mobile App

Production · Google Play

Kotlin · Jetpack Compose · Azure AD · Microsoft Intune · CI/CD

- Developed and shipped the official internal mobility platform for Bangalore International Airport employees, centralizing HCM, shift management, and HR services.
- Built the app in Kotlin/Jetpack Compose; integrated Azure AD single sign-on and Microsoft Intune (MDM/MAM); resolved critical auth blockers to support 24/7 airport operations.

Pulse — Cloud Operations Platform

Arche Global

React.js · Node.js · Express.js · MySQL · AWS · RESTful APIs

- Architected a cloud operations platform for hybrid environment management — infrastructure visibility and incident response.
- Designed 15+ RESTful API endpoints and a MySQL-backed data layer with optimized queries; built real-time monitoring dashboards in React.js.

ArcheOne — Productivity Platform (Android)

Arche Global

Kotlin · Android SDK · RESTful APIs · Push Notifications

- Delivered end-to-end features (Business Card, SOS, Deskart, TravelDesk, Helpdesk) with real-time push notifications and optimized REST integrations across devices.

EDUCATION

Bachelor of Technology (B.Tech), Engineering

2021

Sri Venkateswara College of Engineering, Chennai

Advanced Generative and Agentic AI

In progress · Expected Oct 2026

Naresh Technologies

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate** — valid through March 2027
- **APTECH Certified Python Developer**

LANGUAGES

English (Fluent) · Hindi (Fluent) · Tamil (Native)